

BRENDA ANAGUE

AIMS Research and Innovation
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Education

AIMS RIC (Rwanda) & University of KwaZulu-Natal (South Africa)

PhD, in Data Science

2023-Currently

- **Thesis:** “Physics-Informed Neural Networks Beyond Forward Problems: Source Inversion, Equation Learning, and Convergence Theory”
- **Supervisors:** Dr. Bamdad Hosseini (University of Washington, USA), Dr Issa Karambal (AIMS RIC), Dr Jean Medard Ngnotchouye (University of Kwazulu-Natal, SA)

African Institute for Mathematical Sciences (AIMS)

Limbe, Cameroon

M.Sc in Mathematical Sciences, **Speciality: Data Science**

2021–2022

- **Thesis:** “The Study of Over-parameterization for Physics-Informed Neural Networks”
- **Supervisor:** Dr Yunan Yang, (Cornell University)

University of Dschang (UDs)

Dschang, Cameroon

M.Sc in Applied Mathematics

2019–2021

- **Thesis:** “Valuation and Hedging of Options in Discrete Markets”
- **Supervisor:** Pr. Gabriel Deugoué (UDs)

University of Dschang (UDs)

Dschang, Cameroon

B.Sc in Mathematics and Computer Science,

2016–2019

- **Thesis:** “Mathematical Modelling of Complex Biological Systems”
- **Supervisor:** Dr. Jean Baptiste Patenou, (UDS)

Research Interest

- Deep Learning , Physics Informed Machine Learning, optimization
- Numerical analysis for PDEs.
- Convergence analysis of PINNs using Barron spaces
- Conditional stability based on Carleman estimates for source inversion problems
- Equation learning of PDEs from sparse measurements
- Numerical solution of PDEs using Gaussian Processes

Scholarships, Awards and Prizes

- **Poster Prize Deep Learning Indaba (Rwanda)** Aug 2025
Poster prize, awarded by Nvidia
- **Young African AI Research (YAAIR) Prize Winner, at Deep Learning Indaba Conference (Senegal)** Sep 2024
Poster prize, awarded by InstaDeep
- **Poster Prize Deep Learning Indaba (Senegal): Women In Machine Learning(WIML) Mentorship** Sep 2024

Fully sponsored by CISPA

- **Third place at the Senegalese Serpent Venom Sequence Hackathon: Prize at Deep Learning Indaba Conference (Senegal)** Sep 2024
1000 usd Awarded by InstaDeep
- **PhD fellowship: AIMS RIC & DeepMind** Nov 2022 - Nov 2026
Fully funded
- **Scholarship: African Institute for Mathematical Sciences (AIMS) Cameroon** 2021–2022
Fully funded scholarship
- **President's excellence prize for graduate and postgraduates students** 2019,2020,2021,2022

Prior Research Experience

- **Internship project (InstaDeep)** Dec 2024–April 2025
Retrieval of biological and chemical crop traits from hyperspectral and multispectral data using PINNs
Advisor: Ibrahim salihu Yusuf
- **Master's Thesis (AIMS Cameroon)** April 2022–June 2022
The understanding of the training dynamic of PINNs under an over-parameterized regime.
Supervisor: Dr Yunan Yang, Cornell University
- **First Master's Thesis (University of Dschang)** March 2021–July 2021
Build an investment strategy that allows a seller and a buyer to hedge against financial risks. This project relied on the use of Cox-Ross Rubinstein model and stochastic differential equations such as Black Scholes model.
Supervisor: Prof. Gabriel Deugoue, Univeristy of Dschang
- **Project on Finite Element Methods and Advanced Simulation** 04 Oct. 2020 – 10 Jan. 2021
Finite element methods for Elliptic problems and simulation Supervisor: Dr Laurent Tchoualag, University of Dschang.
- **Project on Partial Differential Equations** 01 Oct. 2020 – 10 Jan. 2021
Hamilton-Jacobi Equation. Supervised by Prof. Jean Louis Woukeng, University of Dschang.
- **Project on Variational Methods and Optimization** 12 Oct. 2019 – 05 Jan. 2020
Sobolev Spaces and the Variational Formulation of Elliptic Boundary Value Problems in N Dimensions.
Supervisor: Prof. Calvin Tadmon, University of Dschang.
- **Bachelor's Thesis** March 2019–July 2019
Use of kinetic theory to build a biological model. Supervisor: Dr. Jean Baptiste Patenou, University of Dschang

Talks and Academic Events Attended

- **Deep Learning Indaba Conference (In person)** 17 Aug 2025 - 22 Aug 2025
Kigali, Rwanda
- **Workshop on TorchPhysics: Deep Learning for PDEs with TorchPhysics (In person)** 14 Nov 2024 - 15 Nov 2024
Werner-von-Siemens Centre Berlin, Germany
- **Deep Learning Indaba Conference (In person)** 01 Sep 2024 - 08 Sep 2024
Dakar, Senegal
- **Data Science Africa Summer School and Workshop (In person)** 03 June 2024 - 07 June 2024
Nyeri, Kenya
- **Workshop: From Theory to Practice in Data Science. Talk on Physics-informed neural networks for atmospheric inverse problem (In person)** 15 April 2024 - 19 April 2024
Organizers: AIMS RIC, Kigali, Rwanda

- **Workshop on AI in Science (In person)** 05 Feb 2024 - 07 Feb 2024
Organizers: Imperial College London and AIMS RIC in Kigali (Rwanda)
- **Global Fellow Program 'African Data Science Solutions' (In person)** 20 Nov 2023 - 24 Nov 2023
Organizers: AIMS RIC and Imperial College London, in Kigali (Rwanda)
- **Deep Learning Indaba Conference (In person)** 01 Sep 2023 - 10 Sep 2023
DL Indaba, Accra (Ghana)
- **International Conference on Learning Representations (In person)** 01 May 2023 - 05 May 2023
ICLR, Kigali (Rwanda)
- **Workshop on Physics for Machine Learning (In person)** 04 May 2023 - 05 May 2023
ICLR, Kigali (Rwanda)
- **Workshop: From Theory to Practice in Data Science (In person)** 23 Apr 2023 - 27 Apr 2023
Quantum Leap Africa, Kigali, Rwanda
- **Doctoral Training School in Data Science (In person)** 04 April 2023
Quantum Leap Africa, Kigali, Rwanda

Publications, Preprints and Ongoing work

- [1] B. Anague et al., "Convergence analysis of PINNs in solving parabolic PDEs under spectral Barron space regularity. (Ongoing work)", 2026.
- [2] B. Anague et al., "Equation Learning, Source function and Initial Condition Identification of parabolic PDEs from sparse measurements data. (Ongoing work)", 2026.
- [3] B. Anague et al., "Physics-Informed Neural Networks for Joint Source and Parameters Estimation in Advection Diffusion Equations", in *Submitted in Journal of Computers and Mathematics with Applications*, 2025, Preprint <https://arxiv.org/abs/2512.07755>.

Research Visit

- **Center for Industrial Mathematics (ZeTeM)** Nov 2024
Visit to the Technomathematical working group, led by Prof. Dr. Dr. h.c. Peter Maaß, University of Bremen Germany.

Technical Skills

- **Programming languages:** Python, Matlab, C++, Jax, R
- **Numerical analysis of PDEs**
- **PDE Analysis:** Banach spaces, Barron spaces, Sobolev spaces, PDE regularity theory and functional analysis
- **Gaussian Processes for solving PDEs**
- **Inverse problems**

Industry Experience

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| Internship:
InstaDeep | Kigali, Rwanda
01 Dec 2024–30 Apr 2025 |
| Internship:
International Bank of Cameroon for Savings and Credit (BICEC) | Douala, Cameroon
Aug 2019–Sep 2019 |

Languages

- **French:** Native
- **English:** Fluent

REFEREES

- **Dr. Bamdad Hosseini**
Department of Applied Mathematics at University of Washington, USA
Email: *bamdadh@uw.edu*
- **Dr. Issa Karambal**
Senior Researcher for Research and Development and Junior Research Chair in Data Science, AIMS RIC Rwanda
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- **Dr. Jean Medard Ngnotchouye**
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